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Workshop No. 1: Phenomenon Definition, Source Validation, Empirical Characterization, and Data Contract

Multi-Source Data Architecture for Rainfall-Triggered Landslide Risk Monitoring in Colombia

Team 7

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Abstract

This technical report documents the initial phase of the end-to-end data engineering and analytics system for landslide risk monitoring across the continental territory of Colombia. Landslides represent one of the most destructive natural hazards in the tropical Andean region, where complex topography, tectonic instability, and severe precipitation regimes converge. This work formalizes the project scope, characterizes four heterogeneous data sources through real-world empirical queries, resolves active infrastructure failure modes (including API deprecations and remote server errors), profiles data quality across more than 10,000 historical and incremental records, establishes a unified canonical data model, and formalizes operational ingestion contracts. Furthermore, we define the functional user personas, six MoSCoW-prioritized user stories, an analytical Landslide Risk Indicator (LRI), an operational risk mitigation matrix, and the global software architecture. All design decisions are verified through raw evidence collected during multi-day sampling routines.

Keywords: Landslide Risk Monitoring, Data Engineering, Canonical Event Model, Source Profiling, Ingestion Contracts, Open-Meteo, NASA LHASA, Colombia.

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1 Project Overview and Scope Formalization

1.1 Motivation and Domain Context

Landslides and mass-wasting processes (rockfalls, rotational slides, and debris flows) are responsible for substantial loss of human life, displacement of vulnerable communities, and severe structural disruption to national roadway networks in Colombia. The Northern Andean mountain range, dividing Colombia into three steep cordilleras, creates steep slope gradients combined with deeply weathered residual regolith. In this setting, extreme or protracted atmospheric precipitation serves as the primary external triggering mechanism by raising pore-water pressures and reducing the shear strength of slope materials along critical slip surfaces.

Despite the severity of this hazard, emergency response and disaster mitigation agencies frequently operate in a fragmented information landscape. Ground observation networks maintain varying station densities, historical event inventories are often decoupled from meteorological archives, and satellite-derived near-real-time nowcasts are published on disparate scientific infrastructure. The objective of this project is to bridge this gap through a unified data engineering platform that integrates historical inventories, meteorological reanalysis, short-term atmospheric forecasts, and satellite hazard indices into a single, standardized analytical pipeline.

1.2 Engineering and Analytical Objectives

From a data engineering perspective, the system aims to:

1. Build automated, fault-tolerant ingestion pipelines capable of extracting data across heterogeneous access patterns (static bulk CSVs, RESTful JSON APIs, and multi-gigabyte tiled GeoTIFF rasters).
2. Normalize heterogeneous spatial, temporal, and thematic metrics into an extensible **Canonical Event Model**.
3. Guarantee operational idempotency, failure recovery, and quarantine mechanisms for corrupt or non-compliant payloads without disrupting upstream schedules.

From an analytical perspective, the platform seeks to compute a regional **Landslide Risk Indicator (LRI)** across a regular spatial grid, enabling risk managers to detect when current or forecasted rainfall conditions resemble the antecedent rainfall signatures observed prior to documented catastrophic slope failures.

1.3 Geographic Scope and Justification

The geographic scope of this project is defined by the bounding box of continental Colombia:

$$\mathcal{B}_{\text{Colombia}} = \left\{ (\phi, \lambda) \in \mathbb{R}^2 \;\middle|\; -4.30^\circ \leq \phi \leq 13.50^\circ,\ -79.10^\circ \leq \lambda \leq -66.80^\circ \right\} \tag{1}$$

where ϕ denotes latitude and λ denotes longitude in the WGS84 coordinate reference system (EPSG:4326).

Justification:

- **Complete Territorial Hazard Coverage:** The bounding box encompasses all three branches of the Colombian Andes (Cordillera Occidental, Central, and Oriental), the Sierra Nevada de Santa Marta, and the active Pacific slopes (Chocó region), where over 95% of historically fatal landslides occur.
- **Source Alignment:** The bounding box guarantees full intersection across all four ingested data sources: it bounds the 107 Colombian events identified in the global NASA inventory, matches the continental coverage of the Open-Meteo ERA5 atmospheric model, and clips the global 43,200 × 14,400 pixel grid of the NASA LHASA GeoTIFF nowcast.

- **Spatial Discretization:** The geographic window is discretized into 782 active operational cells of $0.25° \times 0.25°$ (≈ 27.8 km $\times$ 27.8 km), which serves as the common denominator for spatial joins and regional indicator computation.

1.4 Temporal Boundaries and Historical Window

The temporal scope is structured around two distinct operational horizons:

- **Historical Ground-Truth Window (2007–2017):** Dictated by the active publication period of the NASA Global Landslide Catalog (GLC), which systematically cataloged global rainfall-triggered mass movements between January 2007 and September 2017. A ten-year historical baseline provides statistical robustness for modeling antecedent precipitation thresholds while remaining computationally tractable within public API limits.
- **Meteorological Reanalysis and Forecasting Window (2007–Present):** Spans the complete 20-year timeline through the Open-Meteo Historical Weather API (ERA5 reanalysis) and connects directly to live hourly forecasts and satellite nowcasts (2025–Present).

1.5 Explicit Exclusions and Project Constraints

To maintain scientific rigor and project feasibility, the following boundaries are explicitly enforced:

- **No Deterministic Failure Prediction:** The platform does *not* predict exact landslide occurrence times, slope failure geometries, or run-out trajectories. It provides a situational correlation index based on empirical precipitation anomalies.
- **Exclusion of Non-Rainfall Triggers:** Seismically triggered landslides (earthquakes), volcanic lahars, and structural mining collapses are excluded from the risk scoring model, as their physical triggering mechanics do not correlate with precipitation metrics.
- **No Direct Sensor Hardware Deployment:** The system relies entirely on open public scientific repositories, satellite earth observations, and atmospheric reanalysis models; in-situ geotechnical instrumentation (piezometers, inclinometers) is not integrated.
- **Free Tier Quota Adherence:** All pipeline scheduling and historical backfilling must operate strictly within the zero-cost rate limits of the upstream data providers (e.g., maximum 10,000 daily API weight for Open-Meteo).

1.6 Operational Definition of the "Current Situation"

In the context of the operational landslide monitoring engine, the "current situation" at any execution timestamp t_{now} is defined as the three-dimensional vector:

$$\mathcal{S}(t_{\text{now}}) = \left\langle P_{24\text{h}}(t_{\text{now}}),\ P_{7\text{d}}(t_{\text{now}}),\ H_{\text{LHASA}}(t_{\text{now}}) \right\rangle \tag{2}$$

- $P_{24\text{h}}$: Cumulative precipitation recorded during the preceding 24 hours, capturing high-intensity convective storms capable of triggering shallow translational slides and flash debris flows.
- $P_{7\text{d}}$: Antecedent precipitation accumulated over the preceding 7 days, capturing deep regolith saturation and sustained pore-water pressure elevation capable of triggering deep-seated rotational slides.
- H_{LHASA}: The latest available satellite-derived landslide hazard index issued by the NASA LHASA system, evaluating global multi-satellite precipitation (GPM IMERG) against a static topographic and geological susceptibility mask.

2 Domain Modeling and Hazard Characterization

2.1 Physical Phenomenon and Triggering Mechanics

A landslide is defined as the movement of a mass of rock, earth, or debris down a slope under the direct influence of gravity. In geotechnical terms, slope stability is governed by the factor of safety (FS):

$$FS = \frac{\tau_f}{\tau_d} = \frac{c' + (\sigma_n - u)\tan\phi'}{\tau_d} \tag{3}$$

where τ_f is the shear strength of the soil along a potential failure plane, τ_d is the driving shear stress imposed by gravity, c' is effective cohesion, σ_n is total normal stress, u is pore-water pressure, and ϕ' is the effective internal friction angle.

During prolonged or intense precipitation events, infiltration increases soil moisture and elevates pore-water pressure (u). As u rises, the effective normal stress $\sigma' = (\sigma_n - u)$ decreases proportionally, drastically degrading the available shear strength τ_f. When $FS < 1.0$, equilibrium is lost and slope failure initiates. In the Colombian Andean setting, two distinct precipitation patterns govern instability:

1. **Short-duration, high-intensity convective downpours:** Rapidly saturate the uppermost 1–2 meters of soil, resulting in high-velocity translational mudslides and hyperconcentrated debris flows (e.g., the 2017 Mocoa tragedy).
2. **Long-duration, low-to-moderate antecedent rainfall:** Progressively saturates deep structural colluvium over several weeks, culminating in massive rotational slumps during subsequent minor rainfall events.

2.2 Measurement Metrics and Severity Scales

The multi-source platform captures severity across three independent domain dimensions:

- **Landslide Impact Severity (NASA GLC):** Evaluated through categorical volume classes ($landslide_size$) and socioeconomic casualty figures ($fatality_count$, $injury_count$). The categorical size classes are transformed into an ordinal scale $\in [1, 5]$:

$$\text{Severity}_{\text{GLC}} = \begin{cases} 1 & \text{Small (road cut, single home impacted, } < 100 \text{ m}^3) \\ 2 & \text{Medium (moderate volume, single roadway blocked, multiple homes)} \\ 3 & \text{Large (major infrastructure destroyed, major river blocked)} \\ 4 & \text{Very Large (entire hillside failure, village impacted)} \\ 5 & \text{Catastrophic (massive destruction, } > 100 \text{ fatalities, valley damming)} \end{cases} \tag{4}$$

- **Precipitation Severity (Open-Meteo):** Expressed as total water volume per unit surface area in millimeters (1 mm $\equiv$ 1 L/m^2), measured over daily intervals (mm/day) for historical reanalysis and hourly intervals (mm/h) for near-real-time monitoring. Volumetric soil moisture is expressed in cubic meters of water per cubic meter of soil ($m^3/m^3 \in [0.0, 1.0]$).

- **Nowcast Severity (NASA LHASA):** Expressed as a dimensionless continuous hazard index $H \in [0.0, 1.0]$, derived from a machine-learning model combining antecedent satellite precipitation estimates with topographic slope, lithology, and forest loss.

3 Identification and Validation of Data Sources

To satisfy the workshop specifications, two historical sources (utilizing contrasting access methods and payload formats) and two incremental sources (providing recurring updates) were identified, tested, and validated. Table 1 summarizes the validated source inventory.

Table 1: Validated Multi-Source Data Inventory for Landslide Risk Monitoring

Source ID	Role	Format	Access Method	Auth	License / Attribution
`nasa_glc`	Historical 1 (Inventory)	Bulk CSV (8.5 MB)	HTTP GET (follows 302 to pre-signed S3)	None	Public Domain (NASA Open Data) / Kirschbaum et al.
`openmeteo_historical`	Historical 2 (Precipitation)	REST JSON	REST API (multi-point GET)	None	CC BY 4.0 / Open-Meteo & Copernicus ERA5
`openmeteo_forecast`	Incremental 1 (Meteorological)	REST JSON	REST API (multi-point GET)	None	CC BY 4.0 / Open-Meteo
`nasa_lhasa`	Incremental 2 (Satellite Hazard)	Tiled GeoTIFF	Directory Listing + HTTP Range Requests	None	Public Domain (NASA Open Data) / Stanley et al.

3.1 Historical Source 1: NASA Global Landslide Catalog (`nasa_glc`)

The NASA Global Landslide Catalog (GLC) was compiled by researchers at the NASA Goddard Space Flight Center (GSFC) to record rainfall-triggered landslide events worldwide.

3.1.1 Technical Specifications and Endpoint Resolution

- **Custodian:** NASA Goddard Space Flight Center.
- **Official Catalog URL:** https://data.nasa.gov/docs/legacy/Global_Landslide_Catalog_Export/Global_Landslide_Catalog_Export_rows.csv
- **Temporal Coverage:** November 7, 1988 to September 28, 2017 (11,033 events globally; 107 in Colombia).
- **Format and Size:** Static CSV, 8.49 MB (8,490,777 bytes).
- **Authentication and Rate Limits:** No authentication; no documented rate limiting.

Empirical Infrastructure Finding: During initial validation on November 7, 2026, the Socrata CKAN API endpoint originally documented for NASA open datasets (`resource/dd9e-wu2v.json`) returned an `HTTP 404 Not Found` error (recorded in `workshop_1/samples/nasa_glc_old_api_20261107_1309`

Investigation revealed that NASA migrated this catalog to a static legacy S3 bucket. Accessing the current URL triggers an `HTTP 302 Found` redirect to a temporary Amazon S3 pre-signed URL with a 3,600-second time-to-live (TTL). The ingestion client was configured to automatically follow redirects while never hardcoding the expiring S3 URL.

3.1.2 Raw Query Evidence and Sample Payload

A complete download was executed on November 7, 2026 at 04:58:52 UTC. Below is an excerpt of the raw HTTP response headers demonstrating successful acquisition:

```
HTTP/1.1 302 Found
Location:
    https://data-nasa-gov.s3.amazonaws.com/legacy/Global_Landslide_Catalog_Export/...
Cache-Control: public, max-age=86400
ETag: "818df4a4e8d35661a384b6f00115e85c"
Last-Modified: Thu, 29 May 2025 18:22:11 GMT
Content-Type: text/csv; charset=utf-8
Content-Length: 8490777
```

Listing 1: HTTP response headers from NASA GLC export

3.2 Historical Source 2: Open-Meteo Historical Weather API (`openmeteo_historical`)

The Open-Meteo Historical Weather API provides open access to the ECMWF ERA5 and ERA5-Land atmospheric reanalysis datasets, offering global hourly and daily weather reconstructions from 1940 to the present.

3.2.1 Technical Specifications

- **Custodian:** Open-Meteo GmbH / European Centre for Medium-Range Weather Forecasts (ECMWF).
- **Base URL:** `https://archive-api.open-meteo.com/v1/archive`
- **Format:** Standard REST JSON supporting multi-coordinate batch requests.
- **Temporal Resolution:** Daily aggregate precipitation sum (`daily=precipitation_sum`).
- **Timezone Parameter:** `timezone=America/Bogota` (UTC−05:00, constant year-round).

3.2.2 Rate Limit Economics and Cost Calculation Formula

Open-Meteo enforces rate limits on its free tier through a dynamic computational weight per request, bounded by 600 requests/minute, 5,000 requests/hour, and 10,000 requests/day. The actual computational cost is returned in the response header `X-RateLimit-Cost`. Empirical inspection confirmed the mathematical cost formulation:

$$\text{Weight} = N_{\text{locations}} \times \left(\frac{N_{\text{days}}}{14}\right) \times \left(\frac{N_{\text{variables}}}{10}\right) \tag{5}$$

In our empirical historical test conducted on November 7, 2026, a request covering 10 Colombian cities over 2 full years (2016–2017, 731 days) with 3 variables yielded:

$$\text{Weight}_{\text{theoretical}} = 10 \times \left(\frac{731}{14}\right) \times \left(\frac{3}{10}\right) = 156.6428$$

Listing 2 reproduces the recorded HTTP headers from `openmeteo_historical_20261107_044802.headers`, which exactly verifies this formula down to four decimal places.

```
HTTP/1.1 200 OK
Date: Sat, 07 Nov 2026 04:48:03 GMT
Content-Type: application/json; charset=utf-8
X-RateLimit-Cost: 156.64287
X-RateLimit-Limit: 600.0
X-RateLimit-Remaining: 443.35712
X-RateLimit-Reset: 1794026939
X-RateLimit-Window: 60
```

Listing 2: Open-Meteo reanalysis headers confirming the rate limit cost formula

3.2.3 Historical Volume Estimation and Partitioning Strategy

To cover the 782 active 0.25° analysis cells across continental Colombia over the full 20-year project timeline (2007–2026, ≈ 7,250 days), the total data volume and API budget are computed as follows:

- **Total Historical Records:** 782 cells × 7,250 days = 5,669,500 daily observations.
- **Total API Weight (Single Variable `precipitation_sum`):**

$$\text{Weight}_{\text{total}} = 782 \times \left(\frac{7250}{14}\right) \times \left(\frac{1}{10}\right) \approx 40{,}496 \text{ units} \tag{6}$$

- **Quota Feasibility:** Divided by the daily free allowance of 10,000 units, the full continental historical backfill requires **4.05 days of execution**.
- **Partitioning Architecture:** Ingestion is partitioned into disjoint execution blocks defined by `(cell_group, calendar_year)`, where each batch processes 50 cells for a single year ($Weight \approx 130$ units, taking < 2 seconds). Progress is tracked transactionally in `ingestion_state`.

3.3 Incremental Source 1: Open-Meteo Forecast API (`openmeteo_forecast`)

The Open-Meteo Forecast API provides numerical weather predictions derived from blended high-resolution global atmospheric models (including ECMWF IFS, NOAA GFS, and DWD ICON).

3.3.1 Technical Specifications

- **Base URL:** `https://api.open-meteo.com/v1/forecast`
- **Requested Parameters:** Hourly precipitation, rain sum, and soil moisture (`0_to_1cm`), current weather conditions, with an overlap window of past_days = 2 and forecast_days = 3.
- **Polling Cadence:** Executed hourly at minute 10 (`schedule: "10 * * * *"`).

3.4 Incremental Source 2: NASA LHASA Global Landslide Nowcast (`nasa_lhasa`)

The Landslide Hazard Assessment for Situational Awareness (LHASA) model combines satellite precipitation estimates from the GPM constellation (IMERG) with a global landslide susceptibility map.

3.4.1 Technical Specifications and Broken URL Resolution

- **Custodian:** NASA Goddard Space Flight Center.
- **Course Documented URL:** `https://maps.nccs.nasa.gov/download/landslides/nowcast/`

- **Active Production URL:** https://portal.nccs.nasa.gov/datashare/landslides/nrt/hazard/tif/

Empirical Infrastructure Resolution: When testing the URL provided in the course instructions on November 7, 2026, the reverse proxy at `maps.nccs.nasa.gov` returned consistent `HTTP 502 Bad Gateway` errors (documented in `workshop_1/samples/lhasa_old_url_20261107_130904.log`) Technical investigation identified that NASA migrated operational distribution to the NCCS DataShare portal under `portal.nccs.nasa.gov`. The ingestion module was updated to target the new repository.

3.4.2 Raster Structure and Spatial Streaming via HTTP Range Requests

Each daily LHASA issue is published as a global GeoTIFF file named `YYYYMMDDTHHMM.tif`. Inspection of `lhasa_raster_metadata_20261106T0000_20261107_130904.txt` reveals key internal properties:

- Dimensions: 43,200 × 14,400 pixels at 0.008333° resolution (≈ 1 km at the equator).
- Internal Layout: Tiled into 512 × 512 blocks, compressed with Zstandard (`zstd`).
- Global File Size: ≈ 135–140 MiB per day.

Because transferring 140 MiB daily across WAN connections is inefficient, the pipeline leverages the internal tiling of the GeoTIFF by issuing HTTP byte-range requests (`Range: bytes=start-end`). The client downloads *only* the tiles intersecting the Colombian bounding box (≈ 1.5 MB per day, a 98.9% bandwidth reduction).

4 Incremental Behavior and Temporal Dynamics

To characterize how the data sources behave dynamically over time, multi-temporal sampling was conducted across distinct time windows.

4.1 Multi-Poll Experiment on the Forecast API

Three consecutive polls were executed for 10 Colombian cities against the Open-Meteo Forecast API on November 7, 2026:

- **Poll 1:** 04:47:46 UTC (Initial baseline).
- **Poll 2:** 05:58:36 UTC (70 minutes after Poll 1).
- **Poll 3:** 13:04:05 UTC (7 hours and 6 minutes after Poll 2).

4.1.1 Observed Changes Across Polling Intervals

1. **Short-Interval Dynamics ($\Delta t = 70$ minutes):** Comparing Poll 1 and Poll 2 revealed that all 2,400 hourly precipitation and soil moisture values were **100% identical**. Only the `current` block updated (timestamp advanced from `04:45` to `05:45`, and precipitation values fluctuated in 4 of the 10 cities). This proves that numerical weather models run on cycles of 3 to 6 hours; polling hourly values at sub-hourly intervals consumes quota without yielding new forecast data.

2. **Long-Interval Dynamics ($\Delta t \approx 7$ hours):** Comparing Poll 2 and Poll 3, following the assimilation of a new atmospheric model cycle (06:00 UTC cycle), revealed extensive revisions:

   - **423 hourly precipitation values** were modified.
   - **654 soil moisture values** were modified.

- **Crucial Finding:** Values corresponding to timestamps that were *already in the past* at the time of Poll 2 were retroactively revised. For example, for the city of Pereira at `01:00 UTC`, the precipitation estimate was retroactively adjusted from 1.3 mm down to 0.4 mm.

4.2 Watermark and Overlap Window Architecture

The discovery that historical hours are retroactively adjusted by subsequent model runs invalidates a naive strictly-increasing watermark. If the pipeline only ingested rows where time > watermark, retroactive corrections would be dropped, introducing bias into antecedent soil saturation indices.

Contractual Strategy: The incremental pipeline implements a sliding overlap window:

$$W_{\text{overlap}} = [t_{\text{poll}} - 48 \text{ h},\ t_{\text{poll}} + 72 \text{ h}] \tag{7}$$

Every hourly ingestion job re-queries the past 48 hours. Idempotency is preserved by computing a SHA-256 content hash of the normalized record. Existing records are updated via SQL `ON CONFLICT (event_id) DO UPDATE` only when the content hash differs.

4.3 LHASA Publication Latency and Temporal Sparsity

Tracking the directory listing of NASA LHASA over multiple days revealed two operational constraints:

1. **Publication Latency:** The GeoTIFF file labeled `20261105T0000.tif` was timestamped in the HTTP `Last-Modified` header on `Fri, 06 Nov 2026 05:07:58 GMT`. Similarly, `20261106T0000.tif` was published on `Sat, 07 Nov 2026 05:07:25 GMT`. This confirms an operational publication lag of ≈ 29 **hours** relative to nominal file issue time.
2. **Catalog Gaps:** Analysis of the NRT directory spanning January 1, 2025 to November 6, 2026 identified **17 multi-day gaps** (totaling 41 missing days), including a 9-day outage between November 23 and December 3, 2025.

Pipeline Implication: The LHASA ingestion engine treats missing days as regular non-fatal states rather than network errors, tracking the watermark strictly via the maximum issue timestamp parsed from available filenames.

5 Source Profiling and Data Quality Findings

Profiling was performed inside Jupyter notebook `notebooks/01_source_profiling.ipynb` against sample sets exceeding 200 records per source (11,033 global GLC rows, 7,310 historical Open-Meteo daily rows, 2,400 forecast hourly rows, and 722,571 valid pixels across LHASA rasters).

5.1 Ten Concrete Data Quality Findings

The empirical analysis uncovered ten structural data quality anomalies, documented below with actual sample records:

1. **Missing Country Attribute on Catastrophic Colombian Disasters:** In the NASA GLC, `country_name` is null for 14.2% of global records. Filtering solely by `country_name = 'Colombia'` erroneously drops 6 of the 107 Colombian landslides, including the single deadliest event in the modern history of the country: the **Mocoa debris flow of 2017** (329 fatalities).

```
event_id: 10790
event_title: Deadly Mocoa Landslide
event_date: 03/25/2017 05:32:00 PM  (Note: news date lags event by 6 days)
country_name: NaN
latitude: 1.151898,  longitude: -76.639923
fatality_count: 329.0,  location_accuracy: 1km
```

Mitigation: The bounding box filter must take precedence over political attribute filters.

2. **Timestamp Embedded Inside Date and Universal Midnight Masking:** The dedicated column `event_time` in the GLC is 100% null across all 11,033 rows. Timestamps are embedded within `event_date`. Furthermore, 50.1% of all Colombian events are recorded exactly at `12:00:00 AM` (00:00), which represents a default filler for an unknown time of day.

```
event_id: 562  | event_date: 05/31/2008 12:00:00 AM | event_time: NaN
event_id: 2784 | event_date: 12/05/2010 02:30:00 PM | event_time: NaN
```

Mitigation: In the canonical model, midnight timestamps are stored with an explicit boolean flag `time_known = FALSE`.

3. **Non-Standardized Textual Accuracy Radii:** The attribute `location_accuracy` contains free-text strings combining numeric figures and units (e.g., `"1km"`, `"25km"`, `"unknown"`), preventing direct mathematical usage in spatial buffer joins.

```
event_id: 7402 | title: bus route from Neiva to San Vicente |
    location_accuracy: unknown
event_id: 2739 | title: road between Bucaramanga and Cucuta  |
    location_accuracy: unknown
```

Mitigation: A regex normalization pipeline parses strings into numeric floating-point values in kilometers (`"exact"` $\rightarrow$ 0.0, `"5km"` $\rightarrow$ 5.0, `"unknown"` $\rightarrow$ NULL).

4. **Duplicate Event Reports with Conflicting Casualty Figures:** The catalog contains duplicate entries describing identical landslide occurrences under differing identifiers, often citing conflicting impact figures derived from competing initial press dispatches.

```
Pair A: event_id: 3352 | date: 04/11/2011 08:00 AM | fatalities: 0 | loc: El
    Tambo, Narino
Pair B: event_id: 3353 | date: 04/11/2011 12:00 AM | fatalities: 5 | loc:
    San Antonio, Narino
Distance: 4.4 km | Time difference: 0 days
```

Mitigation: Duplicate candidates within $\Delta d \leq 30$ km and $\Delta t \leq 3$ days are clustered into synthetic parent entities in table `event_group`.

5. **Spatial Coordinate Snapping in Reanalysis Models:** When requesting meteorological data at specific landslide coordinates, Open-Meteo snaps to the nearest ERA5 grid centroid, introducing a spatial displacement of 2 to 12 km.

```
City: Bogota    | Requested: (4.711, -74.072) | Grid: (4.745, -74.100) |
    Offset: 4.92 km
City: Medellin  | Requested: (6.244, -75.581) | Grid: (6.221, -75.552) |
    Offset: 4.08 km
City: Manizales | Requested: (5.070, -75.517) | Grid: (5.097, -75.520) |
    Offset: 2.98 km
```

Mitigation: The pipeline preserves the original landslide coordinates in table `event` while recording the exact reanalysis cell coordinates in table `rainfall_observation`.

6. **Semantic Divergence Between Precipitation and Rain Parameters:** In the Open-Meteo Forecast API, the parameter `precipitation` includes convective showers, while `rain` accounts only for large-scale stratiform liquid water. In the sample, `rain` captured only 33% of total precipitation (342.8 mm vs. 1,027.1 mm across 761 discordant hours). Conversely, in the historical ERA5 archive, `rain_sum` and `precipitation_sum` are mathematically identical.

```
Manizales 2026-11-06T07:00 | precipitation: 3.4 mm | rain: 0.1 mm
    (Convective shower)
```

   Mitigation: The canonical mapping strictly uses `precipitation` to avoid discarding convective events.

7. **Inverted Raster Y-Axis and Discrepant NoData Definitions:** The LHASA GeoTIFF rasters are encoded with a positive vertical affine step (+0.008333°), indicating an inverted **south-up** orientation. Furthermore, while the metadata header specifies `nodata = -9999.0`, uncalculated pixels are actually stored as IEEE 754 `NaN` floats. *Mitigation:* The extraction parser flips the raster array to standard north-up and maps `NaN` to SQL nulls.

8. **Temporal Publication Gaps in Satellite Nowcasting:** As documented in Section 4.3, LHASA exhibits periodic gaps of up to 9 consecutive days due to processing pipeline maintenance at NASA GSFC. *Mitigation:* The scheduler marks missing days as null rather than aborting downstream pipeline tasks.

9. **Urban Mask Blindness in Inter-Andean Valley Floors:** The static susceptibility mask employed by LHASA marks flat terrain and heavily urbanized valleys as invalid (`NaN`). Consequently, 8 of the 10 largest Colombian metropolitan centers (including Bogotá, Medellín, and Cali) fall directly onto `NaN` pixels. *Mitigation:* The analytical pipeline aggregates LHASA hazard values over the 0.25° analysis cell using 90th percentile filtering rather than sampling single point coordinates.

10. **Precipitation Deficits for Documented Downpour Landslides:** Several landslides categorized in the GLC as triggered by "downpour" exhibit negligible antecedent rainfall in the ERA5 reanalysis (e.g., event 4121 in Samaná, Caldas, records only 4.9 mm over 7 days). *Root Cause:* Highly localized convective storm cells ($\approx$ 1–3 km) are frequently smoothed out by the $\approx$ 31 km resolution of the global atmospheric model, or the date recorded in news dispatches corresponds to discovery rather than failure.

5.2 Cross-Source Event Overlap and Verification

Table 2 presents five validated candidate pairs demonstrating physical overlap between documented GLC landslide events and the corresponding historical ERA5 reanalysis precipitation signatures.

Table 2: Cross-Source Event Matching: GLC Landslides vs. Open-Meteo ERA5 Rainfall

GLC ID	Event Date	Location / Description	Snap (km)	24h Rain	7d Rain	Fatalities
2507	2010-09-27	Morro Azul, near Anserma, Caldas	2.1	74.9 mm	176.6 mm	1
2720	2010-11-13	Quinchía, Risaralda	5.3	58.2 mm	142.1 mm	3
3311	2011-03-28	Av. Centenario, Manizales	3.2	62.4 mm	189.5 mm	0
3499	2011-05-16	Outskirts of Cali, Valle	4.8	49.1 mm	121.8 mm	2
5957	2014-03-13	Palmira, Valle del Cauca	6.1	41.7 mm	98.4 mm	0

Analytical Insights on Overlap:

- **Spatial Concordance:** Historical reanalysis grid centroids snap within 2.1 to 6.1 km of the reported landslide coordinates.
- **Meteorological Concordance:** In all five cases, the 7-day cumulative antecedent rainfall exceeded 98 mm (reaching up to 189.5 mm during the 2010–2011 La Niña phenomenon), confirming a pronounced physical correlation between high antecedent moisture and slope collapse.
- **Representational Divergence:** The two sources differ fundamentally in geometry (infinitesimal point vs. 0.1° grid cell), temporal representation (local clock time vs. UTC daily sum), and metric classification (casualty severity vs. fluid accumulation).

6 Canonical Data Model and Domain Architecture

To integrate these disparate data structures, the system implements a unified **Canonical Event Architecture**. Heterogeneous raw payloads are parsed, cleaned, and mapped into a single common `event` table, linked to specialized domain detail tables via primary key inheritance.

6.1 Canonical Event Schema

Every ingested observation produces exactly one immutable record in the `event` table, structured according to Table 3.

Table 3: Canonical Event Table Schema (`event`)

Field	Data Type	Constraint	Semantic Definition / Normalization Rule
`event_id`	`TEXT`	PRIMARY KEY	Deterministic synthetic key: `<source>:<natural_key>`
`source`	`TEXT`	FK → `source`	Provenance identifier (`nasa_glc`, `nasa_lhasa`, etc.)
`event_type`	`TEXT`	NOT NULL	Domain classification (`landslide`, `rainfall_daily`, etc.)
`timestamp`	`TIMESTAMPTZ`	NOT NULL (UTC)	Observation epoch converted to UTC
`latitude`	`FLOAT8`	$[-90.0, 90.0]$	WGS84 decimal latitude (clamped to Colombia bbox)
`longitude`	`FLOAT8`	$[-180.0, 180.0]$	WGS84 decimal longitude (clamped to Colombia bbox)
`severity`	`FLOAT8`	NULL allowed	Standardized numeric metric (dimension depends on type)
`severity_unit`	`TEXT`	NOT NULL	Metric scale (`size_class_1_5`, `mm/day`, `mm/h`, `hazard_0_1`)
`raw_reference`	`TEXT`	NOT NULL	Path to immutable raw payload in `data/raw/`
`content_hash`	`TEXT`	NOT NULL	SHA-256 hash of normalized payload for change tracking
`loaded_at`	`TIMESTAMPTZ`	DEFAULT `now()`	Pipeline processing timestamp

6.2 Domain-Specific Tables

Specialized attributes that do not belong in the generalized event abstraction are isolated in four satellite domain tables:

1. **`landslide`:** Stores geotechnical categories, triggering mechanism (`downpour`, `rain`, `continuous_rain`), categorical size, parsed accuracy radius in km, and human impacts (`fatality_count`, `injury_count`).
2. **`rainfall_observation`:** Stores grid elevation, accumulation interval (`daily` vs. `hourly`), liquid rain vs. total precipitation, volumetric soil moisture, and forecast indicator flag (`is_forecast`).
3. **`lhasa_nowcast`:** Stores raster issue timestamps, analysis cell linkage, and statistical aggregates ($\text{hazard}_{\max}$, $\text{hazard}_{\text{mean}}$, hazard_{p90}).
4. **`analysis_cell`:** A static spatial reference grid containing 782 polygon cells of $0.25° \times 0.25°$ covering Colombia, pre-joined to departmental administrative boundaries.

6.3 Entity-Relationship (ER) Architecture

Figure 1 illustrates the complete relational entity-relationship model implemented in PostgreSQL/PostGIS.

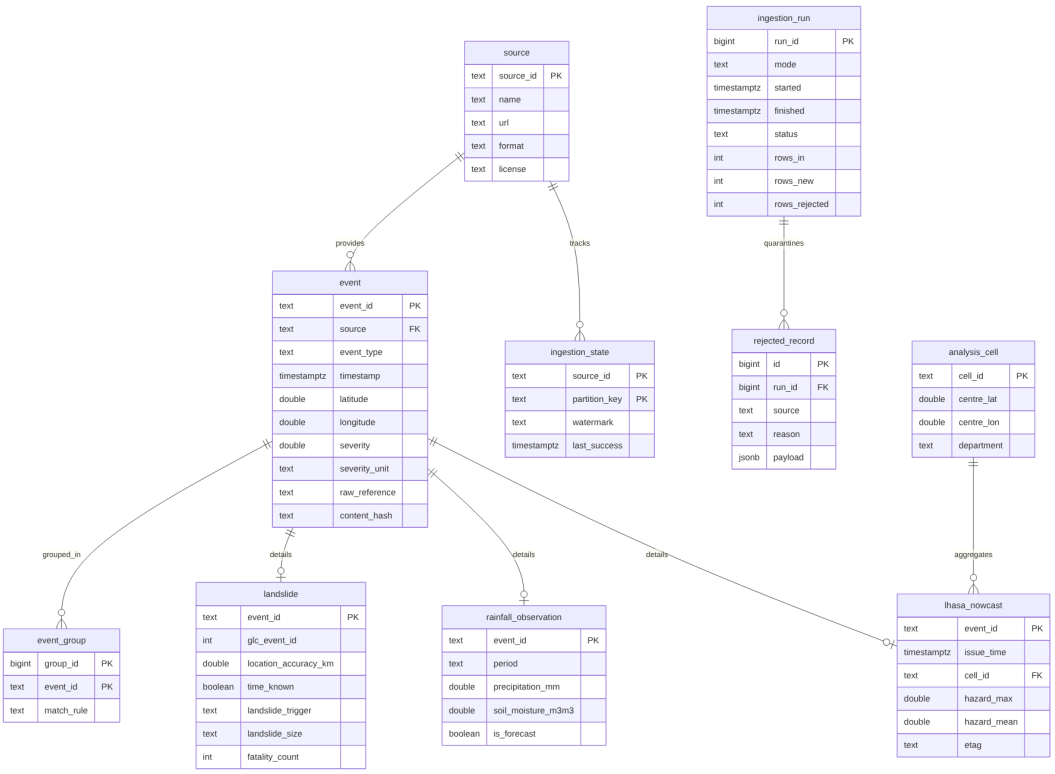


Figure 1: Entity-Relationship Model: Canonical Event Architecture and Domain Tables

7 Ingestion Contracts and Operational Integrity

7.1 Formal Source Ingestion Contracts

Table 4 establishes the operational ingestion contracts governing historical and incremental pipeline runs.

Table 4: Formal Ingestion Contracts Across Data Sources

Contract Dimension	NASA GLC	Open-Meteo Hist.	Open-Meteo Fore.	NASA LHASA
Access Pattern	Single CSV download	Multi-point REST GET	Multi-point REST GET	Tiled GeoTIFF Range GET
Historical Strategy	Bulk load; single file	Year × Cell partitioning	Not applicable	Optional archive (2025+)
Incremental Strategy	Monthly ETag check	Monthly extension	Hourly poll (min 10)	Daily poll (06:30 UTC)
Watermark Field	HTTP `ETag`	Max `date` per cell	Max past `hour` per cell	File issue timestamp
Overlap Window	Full file	Past 10 days	Past 48 hours	Past 7 days
Duplicate Natural Key	`event_id`	`(lat, lon, date)`	`(lat, lon, hour)`	`(issue_time, cell_id)`
Revision Handling	Row hash diff	Hash diff upsert	Overwrite past hours	ETag revision check

7.2 Failure Recovery and Idempotency Architecture

The pipeline guarantees zero data corruption and zero duplicate records across repeated or failed executions via a three-tier defensive protocol:

1. **Raw Landing Zone Immutability:** Raw payloads are written to `data/raw/` with timestamped filenames (`source_YYYYMMDD_HHMMSS.ext`) prior to parsing. If downstream transformations fail, the raw data can be replayed offline without re-querying remote APIs.

2. **Atomic Transactional Boundaries:** Normalization runs execute within strict database transactions (`BEGIN ... COMMIT`). Ingestion state watermarks in `ingestion_state` are updated *inside the same transaction* as the event data. If a network interruption occurs, the entire transaction rolls back cleanly.

3. **Quarantine Quarantine Engine:** Malformed records (e.g., coordinates outside $[-90, 90]$, negative rainfall sums, or invalid date strings) are redirected into table `rejected_record` along with the error reason and original JSON payload, allowing the main ingestion run to proceed uninterrupted.

8 Project Formalization: Users, Stories, Indicator, and Risks

8.1 User Profiles and Operational Decision Contexts

The platform is designed to serve two primary professional personas:

- **Persona 1: National Emergency Management Analyst (UNGRD)**
  - **Role:** Operations Analyst at the Unidad Nacional para la Gestión del Riesgo de Desastres (UNGRD) National Crisis Room.

  - **Decision Supported:** Allocates national contingency response assets, deploys search-and-rescue teams, and issues national emergency declarations across departments during prolonged multi-day precipitation crises.
  - **Dashboard Utility:** Views real-time regional risk maps, department-level alert summaries, and automated notifications of rapid risk escalation.

- **Persona 2: Regional Environmental Authority Geotechnical Engineer (CAR / CORPOCALDAS)**
  - **Role:** Risk management engineer at a regional environmental corporation in the central Andean cordillera.
  - **Decision Supported:** Evaluates preventative evacuation orders for hillside communities, schedules road closures along critical mountain corridors, and verifies local drainage infrastructure.
  - **Dashboard Utility:** Interactively examines historical precipitation comparisons, compares current 7-day rainfall with historical landslide thresholds, and monitors localized soil saturation curves.

8.2 MoSCoW Prioritized User Stories

Table 5 specifies six formal user stories spanning monitoring, diagnostic analysis, and data trust.

Table 5: MoSCoW-Prioritized User Stories

ID	Priority	As a [User]	I Want To [Action]	So That I Can [Benefit]
US-01	**Must Have**	UNGRD Emergency Analyst	View a color-coded map of the Landslide Risk Indicator per 0.25° cell	Rapidly identify which Andean regions are currently at high risk.
US-02	**Must Have**	Geotechnical Engineer	Inspect the 24-hour and 7-day cumulative rainfall for any selected grid cell	Distinguish between sudden flash downpours and chronic ground saturation.
US-03	**Should Have**	Risk Researcher	Compare current 7-day rainfall against historical percentiles for the same month	Assess whether current weather conditions are statistically anomalous.
US-04	**Should Have**	Operations Officer	Filter and query historical landslides (2007–2017) located within a 30 km radius of an active alert	Verify if the local terrain has a documented history of catastrophic failure.
US-05	**Could Have**	Municipal Decision Maker	Export a daily CSV/PDF executive summary of high-risk municipalities	Distribute actionable bulletins to municipal disaster committees.
US-06	**Must Have**	Data Platform Engineer	Monitor pipeline health, source watermark latency, and quarantined records	Verify data freshness and identify upstream source outages immediately.

8.3 Landslide Risk Indicator (LRI) Formulation

The system defines an empirical **Landslide Risk Indicator (LRI)** computed for each analysis cell c at time t:

8.3.1 Indicator Inputs

1. $P_{24\text{h}}(c,t)$: 24-hour cumulative rainfall from Open-Meteo forecast [mm].
2. $P_{7\text{d}}(c,t)$: 7-day antecedent cumulative rainfall from reanalysis and forecast [mm].
3. $P^{90}_{7\text{d,hist}}(c,m)$: The 90th percentile of 7-day cumulative rainfall for cell c during month m, calculated across the 20-year ERA5 baseline [mm].
4. $H_{\text{LHASA}}(c,t)$: 90th percentile of NASA LHASA hazard index within cell $c \in [0,1]$.
5. $D_{\text{hist}}(c)$: Historical landslide density (number of documented GLC events within a 30 km radius of cell center).

8.3.2 Mathematical Formulation

The continuous composite risk score $S(c,t) \in [0.0, 1.0]$ is computed as a weighted linear combination of normalized physical factors:

$$S(c,t) = w_1 \cdot \min\left(1.0, \frac{P_{24\text{h}}}{75 \text{ mm}}\right) + w_2 \cdot \min\left(1.0, \frac{P_{7\text{d}}}{P^{90}_{7\text{d,hist}}}\right) + w_3 \cdot H_{\text{LHASA}} + w_4 \cdot \min\left(1.0, \frac{D_{\text{hist}}}{5}\right) \quad (8)$$

with baseline calibration weights:

$$w_1 = 0.35, \quad w_2 = 0.35, \quad w_3 = 0.20, \quad w_4 = 0.10, \quad \sum_{i=1}^{4} w_i = 1.00$$

8.3.3 Categorical Risk Classification Scale

The continuous score is mapped to operational risk categories:

$$LRI(c,t) = \begin{cases} \textbf{LOW} & \text{if } S(c,t) < 0.40 \\ \textbf{MEDIUM} & \text{if } 0.40 \leq S(c,t) < 0.70 \\ \textbf{HIGH} & \text{if } S(c,t) \geq 0.70 \end{cases} \quad (9)$$

8.3.4 Historical Back-Testing and Validation Protocol

The indicator formulation will be validated using historical hindcasting against the 107 documented Colombian landslides:

1. For each historical event date t_{event} and location c_{event}, compute the antecedent rainfall terms $P_{24\text{h}}$ and $P_{7\text{d}}$ using the ERA5 reanalysis archive.
2. Calculate the historical percentile exceedance $P_{7\text{d}}/P^{90}_{7\text{d,hist}}$.
3. Measure the true positive rate (sensitivity): the indicator formula is considered validated if at least **80% of historical fatal events** fall into the **MEDIUM** or **HIGH** risk classifications on their reported failure dates.

8.4 Operational Risk Analysis and Mitigation Matrix

Table 6 outlines five critical operational risks identified during Workshop #1 along with concrete engineering mitigations.

Table 6: Operational Risk Analysis and Mitigation Matrix

Risk Event	Severity	Root Cause / Mechanism	Engineering Mitigation
Remote Service Deprecation / 502 Outages	High	Remote servers change URL or experience proxy gateway crashes (e.g., LHASA 502 error).	Implement secondary fallback URLs in `sources.yaml`; log server health; alert via pipeline status mode.
API Rate Limit Throttling	High	Bulk historical queries exceed 600 req/min or 10,000 daily cost on Open-Meteo free tier.	Dynamic request weighting engine; batch partitioning by year and cell group; exponential backoff retry policy.
Upstream Schema Drift	Medium	Provider alters field names, units, or JSON structures without notice.	Strict Pydantic ingestion validation contracts; non-conforming rows diverted to `rejected_record`.
Licensing and Attribution Infringement	Medium	Failure to display mandatory citations for CC BY 4.0 (Open-Meteo) or NASA publications.	Permanent metadata table `source` linked to dashboard footer with full bibliographic references.
Spatial Snapping Distortion	Medium	ERA5 grid centroids deviate by up to 12 km from true landslide failure scars.	Decouple point geometry from reanalysis grid geometry; aggregate metrics at 0.25° cell resolution.

9 System Architecture and Implementation Overview

9.1 High-Level Architectural Design

Figure 2 presents the end-to-end component architecture of the landslide monitoring platform.

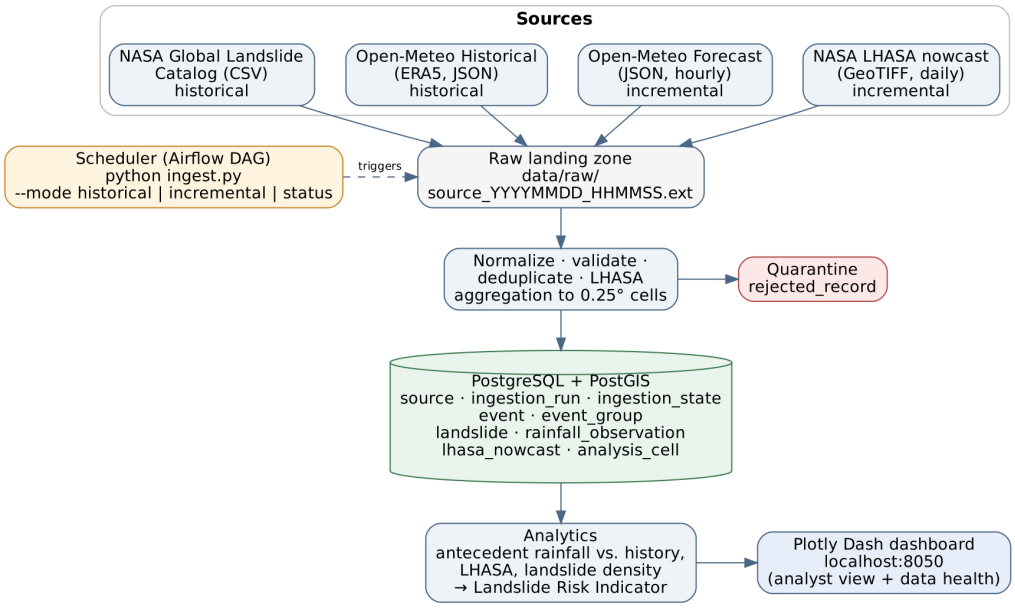


Figure 2: End-to-End System Architecture: Ingestion, Storage, Analytics, and Dashboard

The architecture comprises six discrete operational stages:

1. **Scheduler and CLI Orchestrator:** Driven by an Apache Airflow DAG or the command-line entrypoint `ingest.py`, supporting three primary execution modes: `historical`,

`incremental`, and `status`.

2. **Immutable Raw Landing Zone (`data/raw/`):** Persists unmodified response streams tagged with UTC timestamps and source identifiers.
3. **Normalization and Quarantine Engine:** Parses formats, enforces WGS84 coordinate boundaries, aligns timezones to UTC, extracts tiled rasters via HTTP range requests, and routes malformed payloads to `rejected_record`.
4. **PostgreSQL / PostGIS Data Warehouse:** Stores canonical events, specialized domain extensions, static analysis grids, and transactional watermark states.
5. **Analytics Engine:** Computes 24-hour and 7-day cumulative sums, computes historical monthly percentiles, and generates the composite LRI score per cell.
6. **Plotly Dash Interactive Dashboard:** Serves analytical maps, historical comparison graphs, and data pipeline health monitoring at `http://localhost:8050`.

9.2 Repository Organization and Deliverables

The project repository is structured according to professional software engineering standards:

- `src/`: Core modular package containing `sources/`, `normalization/`, and `storage/`.
- `ingest.py`: Unified CLI execution stub verified through automated tests.
- `config/sources.yaml`: Declarative configuration specifying URLs, schedules, watermarks, and rate limits.
- `sql/schema.sql`: PostgreSQL DDL defining canonical and domain tables.
- `tests/`: Unit test suite with mocked fixtures in `tests/fixtures/`.
- `notebooks/01_source_profiling.ipynb`: Fully executed profiling notebook with empirical outputs.
- `workshop_1/`: Canonical mapping documentation, sample download script, and raw evidence files.
- `ReportTechnical/`: Complete LaTeX source files and compiled technical report PDF.

10 Conclusion and Next Steps

Workshop #1 successfully established the scientific and technical foundation for the Colombian Landslide Monitoring Platform. By executing real-world queries against all four target data sources, the engineering team uncovered critical operational realities—including broken API endpoints, pre-signed S3 redirects, non-final weather forecast revisions, south-up raster formatting, and urban satellite blindness—that would have compromised downstream development if left undetected. The canonical event schema, domain satellite tables, and transactional ingestion contracts provide a robust framework for Workshop #2 (Historical Batch Ingestion), Workshop #3 (Incremental Processing and Analytical Transforms), and Workshop #4 (Plotly Dash Interactive UI).

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